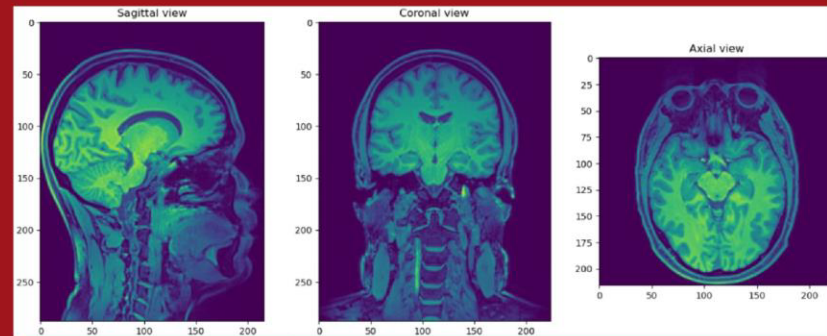


WHAT MATTERS IN THE CEREBELLUM

Obtaining RGB directions of white matter in the cerebellum

OBJECTIVE

The task of the project was to find the structure tensor of the white matter in the cerebellum of the human brain and represent its voxel orientations in RGB color and the scale for each voxel anisotropy.

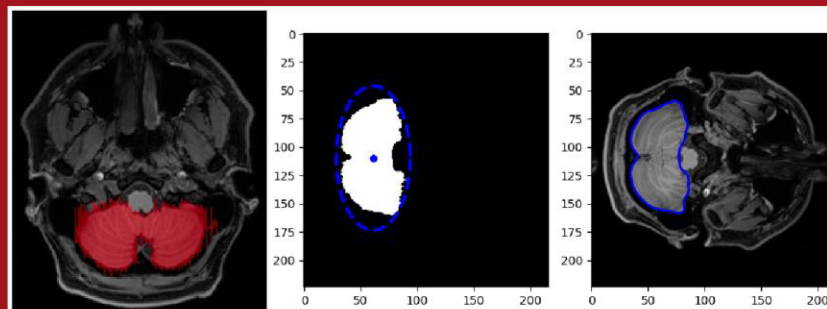


APPROACH

1. Segment out the cerebellum from the brain scan
2. Isolate white matter in cerebellum scans
3. Calculate ST's and voxel anisotropy's
4. Display voxel directions and scales

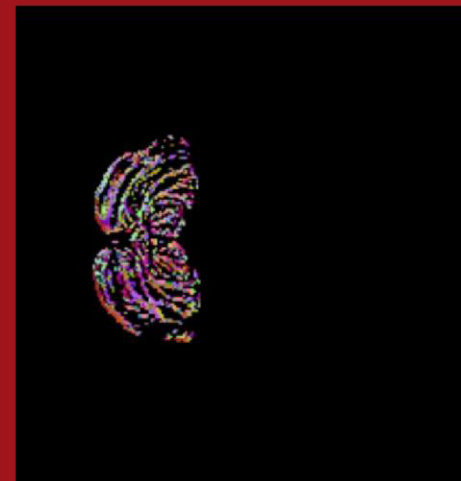
CEREBELLUM SEGMENTATION PRE-PROCESSING

We segmented the cerebellum manually by creating masks using ITK-SNAP, and the cleaning up the segmentation masks using small object removal and binary closing functions. Then the label() function was used to store MRI slices containing the wanted data. The shapes of the masks were used to initialize an ellipse around the center of the cerebellum before an active contour snake was used to isolate the cerebellum, iterating through slices.



RESULTS

The final step was to compute the structure tensors to obtain the directions and scales for the white matter voxels in the cerebellum through fractional anisotropy computation. This was achieved through the 3D structure tensor library that uses sigma to control the scale of measured gradients and rho to control the scale of integration. The result of this provides us with the eigenvalues, representing the scale, and the eigenvectors, representing the orientation of the changes of anisotropy.



Here we see the RGB directions of the anisotropy of the voxels in the white matter of the human cerebellum. This allows us to observe the structural organization and connectivity of white matter tracts within the cerebellum. This could be implemented in medical studies to determine brain health or track neurological changes.



By examining the scale of the anisotropy, medical examiners can examine the organization and possible damage or changes to the white matter. This can again be used to track health problems or observe and study the effects of ageing or neurodegenerative disorders.

DISCUSSION

The task was not complicated in theory, however in practice there was a lot of different processes and methodologies that had to be used in tandem to achieve a result that we could be happy with. One of the hardest parts of the task was to consistently and accurately segment out the white matter of the cerebellum, as there was a large variety of intensities not only between different slices, but also in different parts of the cerebellum in a single slice. We are quite happy with the solution we came up with and implemented and feel as if we have learned a lot from this assignment.

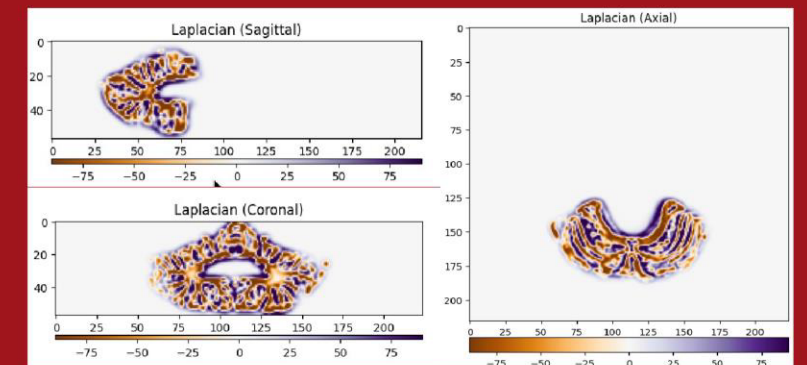
WHITE MATTER ISOLATION

Due to the large variation in voxel intensities in the initial data, we pre-processed the cerebellum scans to enhance the accuracy of our white matter segmentation. We achieved this through adjusting gamma settings and local contrasts, boosting edges and voxel intensities. From there, we implemented FSL FAST segmentation tool to extract the WM as a mask. Finally, we also implemented further vesselness filters to further boost our segmentation accuracy.



SCALE-SPACE ANISOTROPY

To capture local intensity variations in the white matter image, we compute the Laplacian by summing second-order Gaussian derivatives along each principal axis. Using a Gaussian filter with a range of sigma values, the second partial derivatives are estimated, the sum of which yields the Laplacian, highlighting blob-like and tubular structures.



FURTHER WORK

Continuing the task, we would aim to create a more convenient and automated method to segmenting the cerebellum, for example leveraging existing brain segmenting programs which often use deep learning algorithms for higher accuracy. It would also be beneficial to work on the universality and robustness of our program.